SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name Lumina Nusantara

Selected Domain Domain 1 - The Aiversecity Future-Ready Workforce Domain

Selected SDG Track (1–17)

Project / Solution Name SkillRadar MY

Name (Leader) DU YULIN

Email & Phone (Leader) CYS2609009@xmu.edu.my / +60 11-1798 1756

University Affiliation(s) Xiamen University Malaysia (XMUM)

Number of Team Members 5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysian workers and graduates need to understand how changing demand affects their next learning decision. The official SDC Domain 1 brief cites 56% AI-tool adoption against 26% formal training, and a MOSTI estimate of more than 600,000 workers needing reskilling; these are secondary citations from the competition brief, not measurements taken by this team. Static course lists do not explain why a role fits, which skills are missing, or how a recommendation changes when the market assumption changes. We target workers at risk of displacement, fresh graduates, HR and L&D teams, and TVET or career centres who need a transparent starting point for reskilling. The prototype supports career guidance only: it does not predict employment and makes no hiring decision.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillRadar MY accepts a questionnaire or a synthetic text CV. A local keyword and alias parser maps skills onto a team-curated catalogue of 45 roles and 54 skills; it is not a complete O*NET or ESCO identifier mapping. Dated JSON snapshots keep DOSM occupation-level job offerings, JobStreet/Randstad named-skill signals and the WEF global outlook separate, and every panel shows its source, date and scope. A validated manual import updates a snapshot; automatic collection is future work. Five visible scoring components - skill coverage, role focus, transferability, demand trend and gap cost - drive the ranking, and each can be inspected in the interface. Ordered learning steps link to recorded resources, or to a search route where no verified course exists; hours and fees still require confirmation. Role-level and skill-level SDG tags are team interpretations, labelled as such. A moderate and a fast-disruption scenario expose how sensitive the ranking is to assumptions. The application uses Streamlit, Python, JSON and session state, calls no external model API and needs no GPU. Replit screenshots document the development workspace; Streamlit Community Cloud serves the public demo.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Python 3.12 + Streamlit 1.66 for the interface; pandas, NumPy and PyArrow for data handling; Matplotlib for the occupational radar chart. Data lives in versioned JSON files under data/ (roles, skills, courses, dated trend snapshots, copy). No database, no external API and no GPU on the request path. The live demo runs on Streamlit Community Cloud; GitHub hosts the source and the submission folder; Replit is documented as the development workspace. Build-time only: ReportLab generates these PDFs. No SDC sponsor-provided GenAI API or cloud platform is used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The problem definition comes from the Domain 1 brief and public labour-market reporting, not from interviews: this revision makes no claim about user pilots, and an earlier five-person claim has been withdrawn. Design decisions are visible in the product instead: the gap, the five score inputs, the evidence behind each trend, the two scenarios and the learning steps are all on screen, so a reviewer can challenge them without reading code. Deployment follows the same principle: a public repository, a public demo link and a documented local run command. Repository tests and the development log record regression checks; they are not a substitute for independent user validation, which remains future work.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: SDG 4.4 - relevant skills for employment. Secondary: SDG 8.5 - decent work and equal pay for work of equal value. Supporting: 9.5 and 10.2 through skill-level tags.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The Domain 1 brief reports 56% AI-tool adoption against 26% formal training (a 30 percentage-point gap) and more than 600,000 workers needing reskilling within 3-5 years. A product-specific baseline (time to produce one sourced reskilling plan) has not been measured yet; earlier claims of 3-8 weeks are withdrawn.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median minutes for a participant to produce one sourced reskilling plan; relevance rating (1-5) of the top three suggested roles; percentage of displayed recommendations carrying either a cited source or an explicit team-assumption label.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	With at least 10 consenting adult testers on synthetic profiles: median plan completion within 5 minutes, at least 70% rating a top-three role 4/5 or higher, and 100% of recommendations labelled with source or assumption. These are targets, not results.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

Answer: none.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.12 runtime	code	PSF License
Streamlit 1.66.0	code	Apache License 2.0
pandas / NumPy / PyArrow	code	BSD 3-Clause / BSD 3-Clause / Apache 2.0
Matplotlib 3.11.2	code	Matplotlib licence (BSD-style)
ReportLab (build-time PDF generation)	code	BSD 3-Clause
SDC Hackathon submission templates	document template	provided by the organisers

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

Answer: none declared.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: _____ Name: DU YULIN

Date: _____ (to be completed on signing)